

IIT Guwahati

North Guwahati 781039, Assam, India

Techn^{owl}thon

the international school championship
.....Inspiring Young minds!

PAPER THEME



JUNIORS SQUAD



Techniche

The annual Techno-Management Festival
IIT Guwahati

General Instructions

- This question paper consists of 5 sections with a total of 18 questions (as this is the 18th edition of Technothlon).
- There is a bonus section, in the end, you will get to know about it as you go through the question paper.
- The marking scheme is clearly given below and it is one of the most unique schemes you might ever experienced.
- There are two sections (Section 3 and 5) where you have to select the marking scheme according to your choice in the first question (Q.0) of the section. Failing to opt the marking scheme of your choice will result in selection of the marking scheme in the first option by default.
- No ambiguity related to the marking scheme will be entertained later.
- If any discrepancy in the paper is found, the final decision regarding the same will be taken by Team Technothlon itself.
- Integer type answers can have any integer value (NOT restricted to 0 to 9)

Marking Scheme

- **Choose your ally:**

The most interesting marking scheme you might have seen. Out of the given two options; choose the marking scheme that you think is best for you. (You can check the questions before marking your choice)

- **Power sequence:**

Power Scheme: With negative marking

For consecutive correct questions, you will be awarded 20, 21, 22, 23 and so on.

However, if you leave a question or answer it incorrectly, the sequence is broken, and you will start

again from 1.

If you answer the question incorrectly your marks will be deducted in the same sequence

E.g. Solving 4 questions,

For all correct you get $1+2+4+8 = 15$

For all incorrect you get $-1-2-4-8 = -15$

For RRWW you get $1+2-1-2 = 0$

For RRRW you get $1+2+4-1 = 6$

For RRWR you get $1+2-1+1 = 3$

For RRUR you get $1+2+0+1=4$

And so on.

(R -> right answer, W-> wrong answer, U-> Unattempted)

- **Affinity in a row:**

Q1, Q2, Q3 have marks 2, 4 and 6 respectively **BUT**

There are two conditions:

1. There would be no reduction in the marks of this section if you solve 2 consecutive questions correctly, also
2. If the above criteria is not met alongside you solving a question incorrectly the total marks you have scored in this section are halved.

For example :

For a 3 question section-

If you solve Q2 and Q3 correctly but Q1 incorrectly or leave it unattempted then you will get $0 + 4 + 6 = 10$ marks.

If you solve Q1 and Q3 correctly but Q2 incorrectly or you have not attempted it then you will get $(2 + 0 + 6)/2 = 4$ marks.

Similar rules are applied for a section containing 4 questions with the last question having 8 marks.

- **Penaltyless Victory**

Answering a Question correctly Gives You 5 marks but no marks are deducted for incorrect answers. (+5,0)

- **Good ol' Scheme**

Answering a question correctly gives you 5 marks, whereas answering incorrectly reduces 2 from the total marks you gain. (+5,-2)

- **Shield me**

You get 5 marks for answering correctly and 4 marks are deducted for answering incorrectly, but if you have solved the previous question correctly then there is no negative marking for the current question.

For example:

If Your sequence is RWRW then your score is $5+0+5+0 = 10$

But if your sequence is RWWR then your score would be $5+0-4+ 5 = 6$

And if your sequence is RWUW then your score is $5+0+0-4 = 1$

Where (R -> right answer, W-> wrong answer, U-> Unattempted)

- **Risk-Reward**

You get 6 marks for answering any question correctly and none are deducted for an incorrect answer, but if you have answered any question incorrectly your marks for this whole section is halved.

For example: In a section consisting of 4 questions if you solved any two correctly and any one of the other two or both incorrectly you will get 6 marks.

- **All or Nothing**

We have a small surprise for you in the end (well now you will go to the end and check it out and it's no longer a surprise but anyway I have heard Bob saying that sometimes knowing the surprise beforehand is also fun)

You will get full marks in this section only and only if you solve all the questions correctly. No negative marking for any incorrectly solved question.

SECTION 1:

Marking Scheme: Power sequence

Maximum marks: 15

Minimum marks: -15



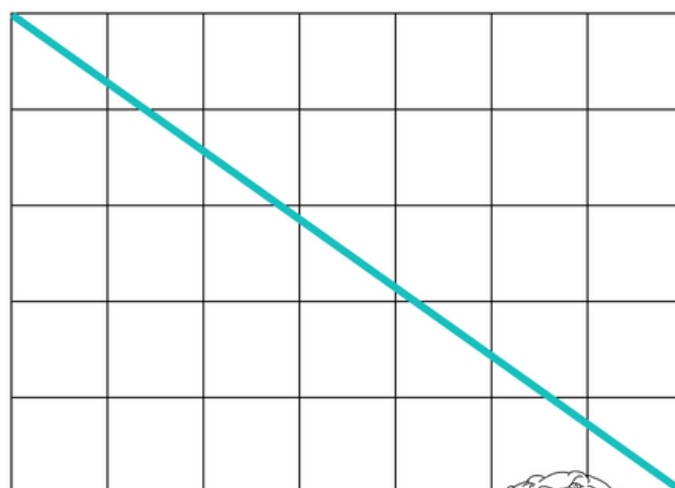
Minions are small, yellow creatures that have existed since the beginning of time, evolving from yellow single-celled organisms into beings that exist only to serve history's most evil masters.

1. [SINGLE CORRECT ANSWER]

In pursuit of finding the evildest master on the earth, minions got the master they were looking for, a Tyrannosaurus Rex. While roaming in the forest they came across a grid. While going through the grid, the T-Rex fell down after it tripped over a stone and started rolling over from one corner of the 224 x 140 grid space in the direction of the volcano, at the diagonally opposite corner. The Tyrannosaurus roars whenever it hits any of the surfaces of the grid. For example, in a 1x2 grid, you will hear 2 roars.

How many total roars will you hear in the case given above?

A small diagram is shown below just for your reference. (It is not according to the measurement of the grid)



- A) 364
- B) 224
- C) 140
- D) 336

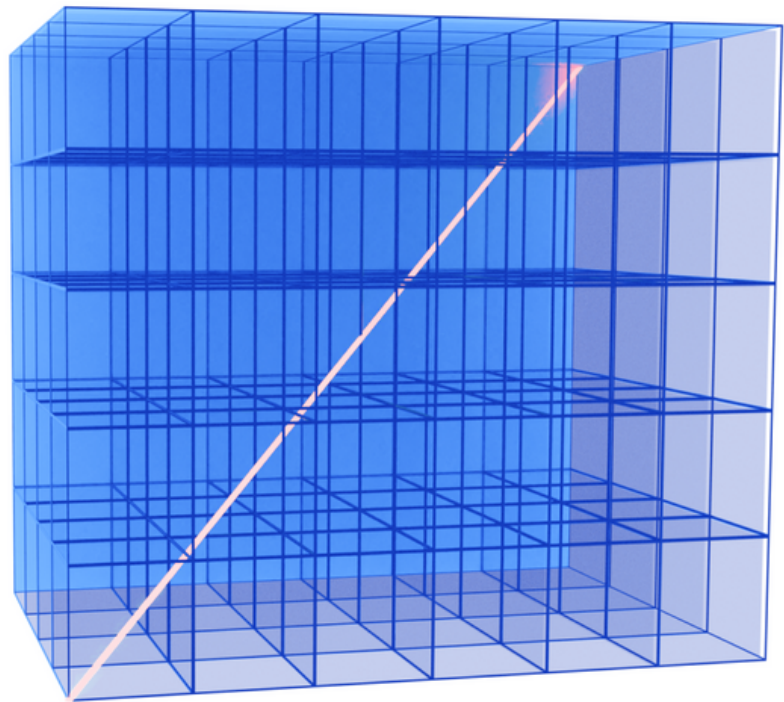
After rolling their leader T-Rex into the volcano, minions were again without a leader. After living in exile for 60 million years, Kevin(the bravest minion out of them all) started moving towards Europe in order to find a new evil leader, and they found the great Napoleon Bonaparte.

2. [SINGLE CORRECT ANSWER]

While practicing his shooting skills, Kevin accidentally fires the bullet in the direction of Napoleon. The bullet entered Napoleon's $1088 \times 574 \times 2041$ palace from one corner and was moving towards Napoleon, standing at the diagonally opposite corner. Napoleon's palace was an architectural wonder; every $1 \times 1 \times 1$ space was covered with walls.

Bullet just failed to reach Napoleon, how many maximum holes it must have created in the journey?

- A) 3703
- B) 3699
- C) 3700
- D) 2041



Fearing Napoleon's wrath, Minions are driven into isolation and start a new life in a massive cave. After many years, the Minions become depressed and unmotivated without a master to serve. To regain their dignity and a sense of purpose, Kevin recruits Stuart and Bob, setting out to look for a new master.

3. [SINGLE CORRECT ANSWER]

Stuart and Bob devise a path to follow. Stuart writes down the first six intervals for a break:
0 1 2 3 4 5

Each subsequent interval is the last digit of the sum of the previous six intervals. Stuart starts scribbling down the entire sequence of intervals.

Help Bob find out which of the following subsequences will never occur.

- A) 1 3 5 7 9
- B) 9 7 5 3 1
- C) 3 1 5 9 7
- D) All

*The three minions move towards New York City, where they end up discovering the **Super-villain Gru**. The three eventually decided to follow GRU to his place.*

4. [MULTIPLE CORRECT ANSWER]

Before letting the minions join his team, GRU decides to test the minions by giving them a challenge. He takes 2 packs of diamonds with x and y diamonds respectively. Gru and Bob, the minion alternatively remove diamonds from one of the packs such that the number of removed diamonds is a multiple of the number of diamonds in the other pack. Bob, the minion makes the first move. The first person to empty a pack wins! Since Bob is very clever, he knows if he picks certain types of packs, he can devise a strategy to win; no matter what moves Gru makes. **For what values of x and y BOB will win the challenge against GRU?**

- A) 177,364
- B) 200,256
- C) 150,402
- D) 21,25

SECTION 2:

Marking Scheme: Affinity in a Row

Maximum marks: 12
Minimum marks: 0

*Super-villain Gru, Impressed by the logical skills of minions, decides to induct them into his team. With his minions, GRU plans to pull off the biggest heist of the century. Gru tries to get a loan from the Bank of Evil and meets a **young super-villain, Vector**. **Bank president Mr. Perkins** tells Gru that he is getting too old and that new super-villains are more youthful and better, like **Vector**.*

1. [INTEGER TYPE ANSWER]

Mr. Perkins refuses to grant **Gru** the loan until he is able to solve the mystery puzzle with a minimum amount of money.

The puzzle is as follows –

There is a 2021x2021 grid and each cell of the grid is either empty or has a token(either 'X' or 'O'). The total number of tokens is 1949. **Young super-villain, Vector** can place the tokens anywhere on the grid and can choose any number of **X** or **O** tokens as long as in total they are 1949.

X	O			X
		X	O	
X		O		
O		O	X	
	X			

A winning configuration has either 3 consecutive **O**'s or 3 consecutive **X**'s in a row or column (and not in diagonal). Any other configuration is a draw configuration. **Mr. Perkins** gives **Gru** 649 free moves and charges \$1857 for each extra move he requires. Gru can in one move flip any token(i.e, Change an **X** to **O** or an **O** to **X**) Finally, he has to make the grid into a draw configuration.

If he uses too much money, he would be at a financial loss and **Mr. Perkins** might not consider him smart enough. So, What is the minimum amount of money he would have to use given **Vector** can place the tokens wherever he wishes.
Calculate the sum of digits of your answer and give that as the final answer.

*Gru and his minions solved the mystery puzzle to get the shrink ray(the ultimate weapon to shrink anything into any size; **FUN FACT:-** GRU used the shrink ray to shrink the whale into the size of a tadpole), but Vector tricked GRU to snatched the shrink ray from him. Gru attempts to get the shrink ray back from Vector's den, but all his attempts to enter the den prove futile.*

*One day, while keeping a watch on Vector's den, he saw three orphaned girls, Margo, Edith, and Agnes, entering the den to sell cookies to Vector.
Gru now wants to adopt those girls from the orphanage, to use them for entering into Vector's den to steal back the shrink ray.*

2. [INTEGER TYPE ANSWER]

Miss Hattie(the head of that orphanage) wants to check GRU's mental ability to ensure if he's stable enough to take care of the three girls.
She gives the Grid* of numbers (as shown below) to Gru.

After looking at the grid, Kevin, who was fond of solving Puzzles from the daily newspapers, said that I have seen such Grids but they only had numbers between 1 and 9, both inclusive. Gru heard the remark and thanked Kevin since he got the most needed clue to solve the question.

Miss Hattie asks him to find the sum of all missing numbers of the grid. Can you figure out the answer?

(*Grid given on the next page)

5	3			49				8
6			1	81	25			
	9	8		16			216	343
64				216				3
16			512		27			1
49				8				6
	216			3		4	64	
			4	1	9			25
				8			49	81

3. [INTEGER TYPE ANSWER]

Miss Hattie then asks Gru if he can keep up the emotional wellbeing of girls.

Gru makes a framework in which he'll use the minions to make girls happy. He placed 8 minions **Stuart, Kevin, Bob, Dave, Phil, Tim, Carl and Jorge**, such that they are not in line with another minion horizontally, vertically, or diagonally. He thought to keep all grey places empty so as to give some private space to the girls.

How many such combinations can Gru make?

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48
49	50	51	52	53	54	55	56
57	58	59	60	61	62	63	64

SECTION 3:

Marking Scheme: Choose your ally

Given Two options choose the marking scheme that is most suitable for you.

Option 1 - Affinity in a row

Option 2 - Good ol' Scheme

Maximum marks: 20

Minimum marks: -8

Q0) [SINGLE CORRECT ANSWER]

So after going through the questions in this section, which marking scheme do you want to choose?

1. Affinity in a row
2. Good ol' Scheme

Gru finally adopts the girls and builds some robots disguised as cookies. He then makes the girls sell those cookie robots to Vector, and these cookie robots help Gru in taking a sneak peek at Vector's den.

1. [SINGLE CORRECT ANSWER]

Vector places **a** cookie robot in the middle of his huge (take it as 200m x 200m) two-dimensional table at his den (consider this middle point as the origin point).

The cookie(robot), deploys many bots to save his time. So all the bots start from the origin. Each second a bot moves exactly 1 meter in one of the four cardinal directions: north, south, west, and east.

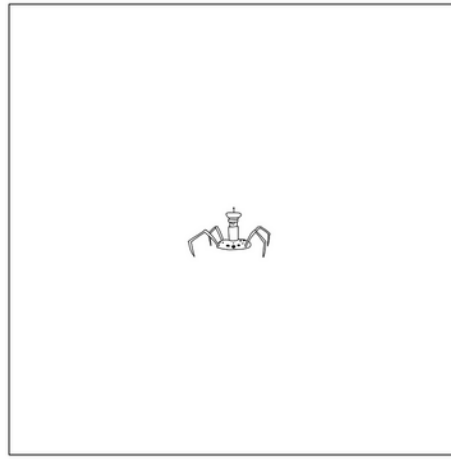
For the first step, the bot **can choose any of the four directions**, but then at the end of every second, it **has to turn** 90 degrees left or right with respect to the direction it just moved in (which is the bot's own choice).

For example, if the bot has just moved north or south, the next step it takes has to be either west or east and vice versa.

Each bot makes **exactly** 100 steps from its starting position according to the rules above. After they reach their destination (final point) the bots are asked to stay as it is. The cookie robot wants the bots to reach a maximum number of different points after the end of 100 steps(every bot travels 100 steps), so minimum, how many bots are required to cover those points.

Note: Two bots or more finally at the same position will make the cookie(robot) angry so bots take care of this. And only the required number of bots are deployed.

- A) 10000
- B) 2500
- C) 2601
- D) 10201



The bots finally helped Cookie(robot) in his mission. As the bots stayed there and scanned that area for any surveillance cameras and disabled them, Gru got a chance to steal the shrink ray gun. The cookie robots help Gru to keep track of Vector's activities. Gru, the girls, and the minions head towards his place.

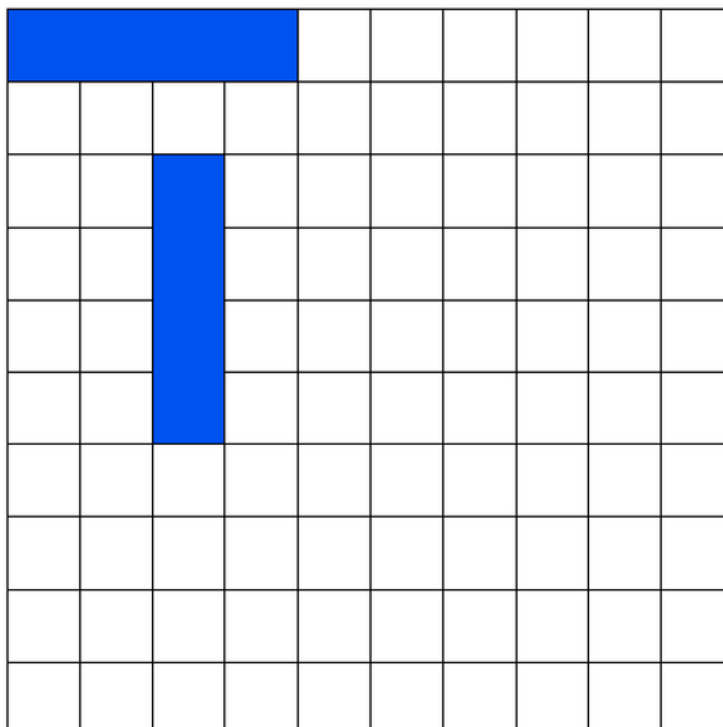
Tired of this rip-roaring but important task, they decide to have some fun. They walk across an amusement park and want to go and play over there.

Gru agrees, intending to leave the girls there. Instead, the attendant says he, as an adult, has to play with them.

2. [INTEGER TYPE ANSWER]

They moved towards a 10x10 square play area and 4x1 tiles were kept. They have to cover the grid using these tiles. Tiles could be placed horizontally or vertically. **Bob**, the minion, helps place the tiles.

But the question is, can you tell in how many ways the grid could have been covered. Report the **sum of digits** of the number of ways as your answer.



3. [INTEGER TYPE ANSWER]

Meanwhile, **Stuart and Kevin** enter a play zone that is of the shape of a circle and has a radius of **D** meters. The play zone has a toy at its center, which is small and easy to lift. **Stuart** and **Kevin** are playful and discover an exciting game. They decide to move the toy but also while keeping it inside the play zone always. A player can move the toy **k** meters towards the east or **k** meters towards the north in one move. The player who can't make a move loses; remember that the toy should always be inside the play zone.

If $D = 56$ and $k = 3$, determine who wins if **Stuart** plays first and they alternate turns and both play optimally. Report the answer as 11 if Stuart wins, 22 if Kevin wins, and 33 if there is no such optimal strategy.

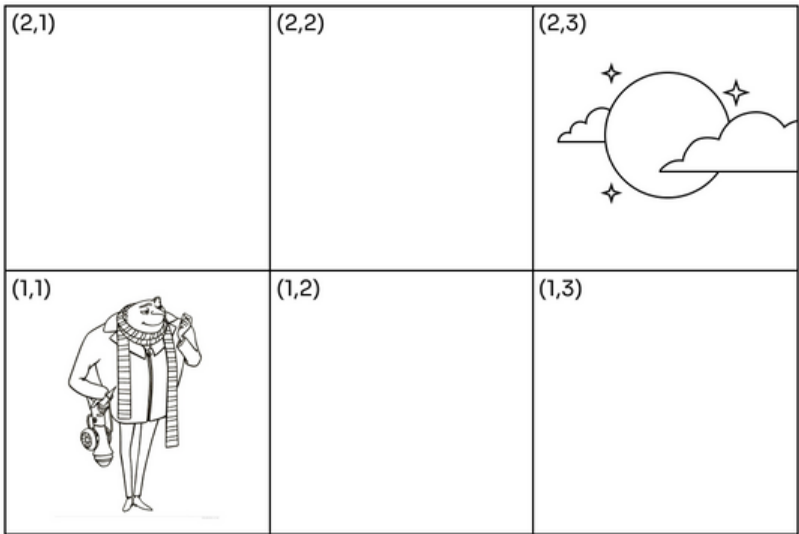
They finally go on a roller coaster ride and then to a snow world park. Everyone enjoyed it a lot and had so much fun. Stuart even ended up pranking Kevin into thinking that he was a girl Kevin liked!

They head to their place after a tiring, thrilling fun-filled day.

Back at home, Gru meets Mr. Perkins. They discuss the shrink ray gun and the further GRU elaborate his evil plan, which is to steal the moon by shrinking it with the help of his shrink ray.

4. [INTEGER TYPE ANSWER]

Gru considered an **nXm** space where he's standing on cell 1X1 (i.e. on the starting cell) has to reach the moon on cell (n,m). From a cell (x,y), he can either move to cell (x+1,y), which costs him y rupees, or to cell (x,y+1), which costs x rupees. There are many different possible paths for reaching cell (n,m) and If he travels along each and every such path, find the sum of all **distinct** amounts of money he paid if $n = 20$ and $m = 30$.



For example in the above figure we have a 2x3 Grid and Gru starts from 1X1 cell and the moon is on the 2X3 cell and he has to reach the moon.

SECTION 4:

Marking Scheme: Penaltyless Victory

Maximum marks: 15

Minimum marks: 0

Perkins again refuses to give him the loan, claiming that although he doesn't have a problem with the plan, he just wants a younger villain to do it instead. Mr. Perkins calls Vector (revealed to be his son) to the Bank of Evil to inform him that Gru has the shrink ray.

Vector reassures his father that he will get the Moon.

1. [INTEGER TYPE ANSWER]

Vector wants to find how a shrink ray gun works, he remembered that there was a scientist named Professor Owlbert in the suburb who knew about this technology.

Vector travels to his lab and asks Prof. Owlbert to tell him about the technology and how it works.

Professor Owlbert doesn't want this unique and dangerous technology to fall into the wrong hands. He wants to test Vector first.

Hence Prof. Owlbert has made a simple setup. He places 4 blue mini bots and 4 red mini bots on the squares shown below.

Then Prof. Owlbert asks Vector to swap the position of all the blue mini bots with red mini bots. The mini-robots can only move vertically or horizontally (just like Rooks/Castle in Chess) and different colored bots (of the same region) should never be in the same horizontal or vertical line

because they would end up destroying each other.

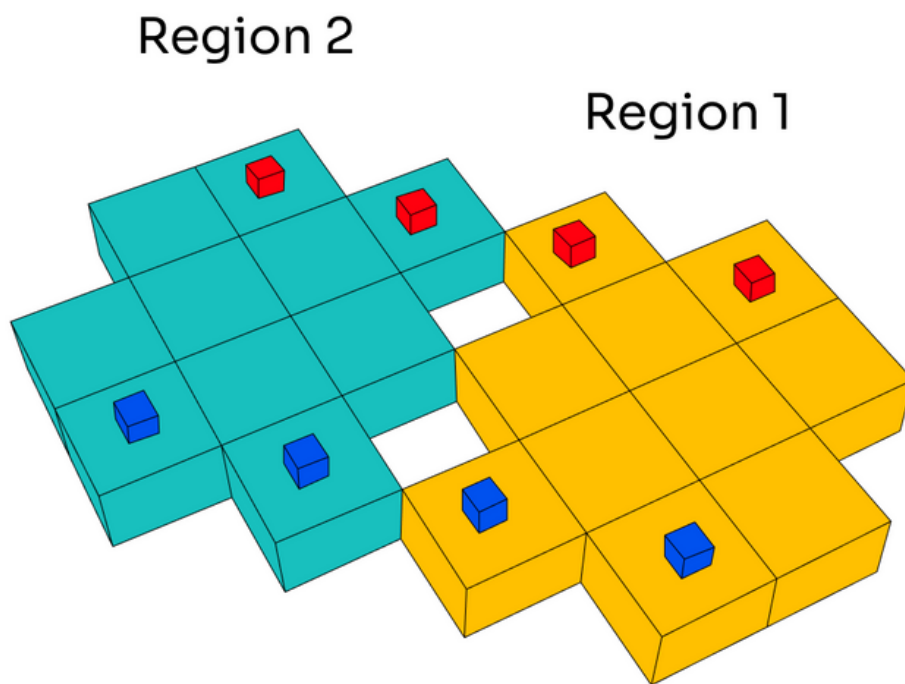
In how many minimum moves can he do this?

Note:

- He has to move Blue and Red bots alternatively on each turn i.e. if he starts with blue first then he has to move red on the next turn.
- A move is played when both the turns are completed.

Eg: If he moves blue first and then after moving red mini-robot a move is counted

Region 1 and Region 2 are independent of each other and mini bots in region 1 don't affect mini bots in region 2 and vice versa.



2. [INTEGER TYPE ANSWER]

Vector then took a bag consisting of n minion models numbered 1 to n to devise the plan. He asked Mr. Perkins to take out any 2 models from the bag with numbers say x and y with the following rules, and add a minion model with the number $x+y+x*y$ into the bag. (He i.e Mr. Perkins can add the model even if $x+y+x*y$ exceeds n . Vector has that stock) Thus after every move the number of minion models decreases by 1 and in the end Vector is left with only 1 minion model.

Let the number written on that model be P . find the sum of all distinct possible values of P for a fixed $n = 9$.

($x*y$ is x multiplied by y)

While Vector struggles to get the shrink ray, Gru almost abandons his plan due to lack of funds, but the girls and the minions take up a collection to keep it going. Gru goes on with his plan and flies to the moon.

Gru successfully shrinks and pockets the Moon. Gru remembers the ballet recital and rushes to it. However, it had ended by the time he arrived. There, he finds a ransom note from Vector demanding the moon in exchange for the girls. Gru hands over the shrunk moon to Vector.

3. [INTEGER TYPE ANSWER]

Vector will let girls go if Gru is able to solve his puzzle. Vector has a robot which gives some output on pressing some switches, vector wants to get the 'truth' statement from that robot, so he assigns the task of tackling the robot to GRU:-

There are infinitely many switches numbered consecutively from 1-7 like6,7,1,2,3,4,5,6,7,1,2,3,4.... And so on connected to a robot, and whenever you press a switch, the robot will display a message which is either a 'true(truth)' or 'false(lie)' statement.

Out of 7 distinct numbered switches, the robot gives a true(truth) statement only when one particular numbered switch is pressed while it gives a false(lie) statement when the other six numbered switches are pressed.

Now GRU has to press switches turn by turn. While pressing the switches, GRU can't see the number written on the switches he presses. On each turn, a switch is pressed and the robot displays a message accordingly.

On the first turn, the robot said: I lie when switches 2 or 3 are pressed.

On the second turn, the robot said: Either of the switches 5, 6, or 7 were pressed during this turn.

On the third turn, the robot said: I lie when the switches 4 or 6 are pressed.

As the switches were hidden beneath the chamber you can't see the number on the switch you are pressing.

But Gru, being smart, pressed three consecutive switches (eg: he might have pressed switch 2,3,4 or 6,7,1, etc.)

Tell the number of that switch which when pressed, makes the robot tell the true statement(truth).

SECTION 5:

Marking Scheme: Choose Your Ally

Option 1 - Shield Me

Option 2 - Risk-Reward

Maximum marks: 24

Minimum marks: -16

QUESTION 0: So after going through the questions in this section, which marking scheme do you want to choose?

- a) Shield Me
- b) Risk- Reward

Vector breaks his promise and keeps the girls and the moon even after GRU solves the puzzle. This enrages Gru, who storms Vector's ship which is conical in shape.

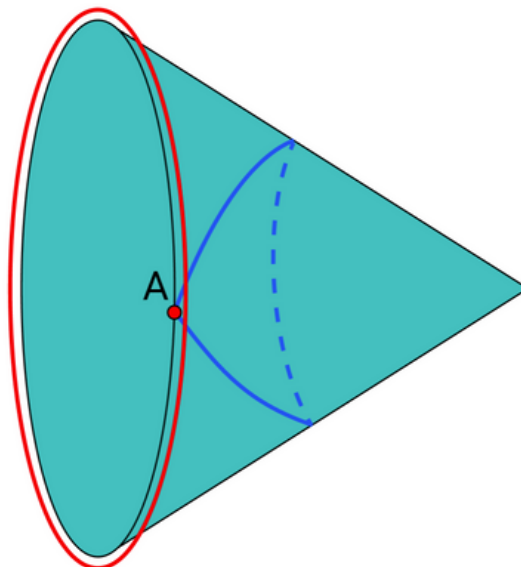
1. [SINGLE CORRECT ANSWER]

Gru started to proceed from point A of the circumference of the circular base of Vector's conical ship. Now he has to make one complete round across the ship through its slanted face. Find the shortest distance he can travel to complete the circle and reach back to point A.

Find the minimum total distance that he might have traveled?

[Take radius and slant height of Vector's ship as $R = 20\text{m}$ and $L = 40\text{ m}$

- a) 40m
- b) 80m
- c) 125.66m
- d) 75m



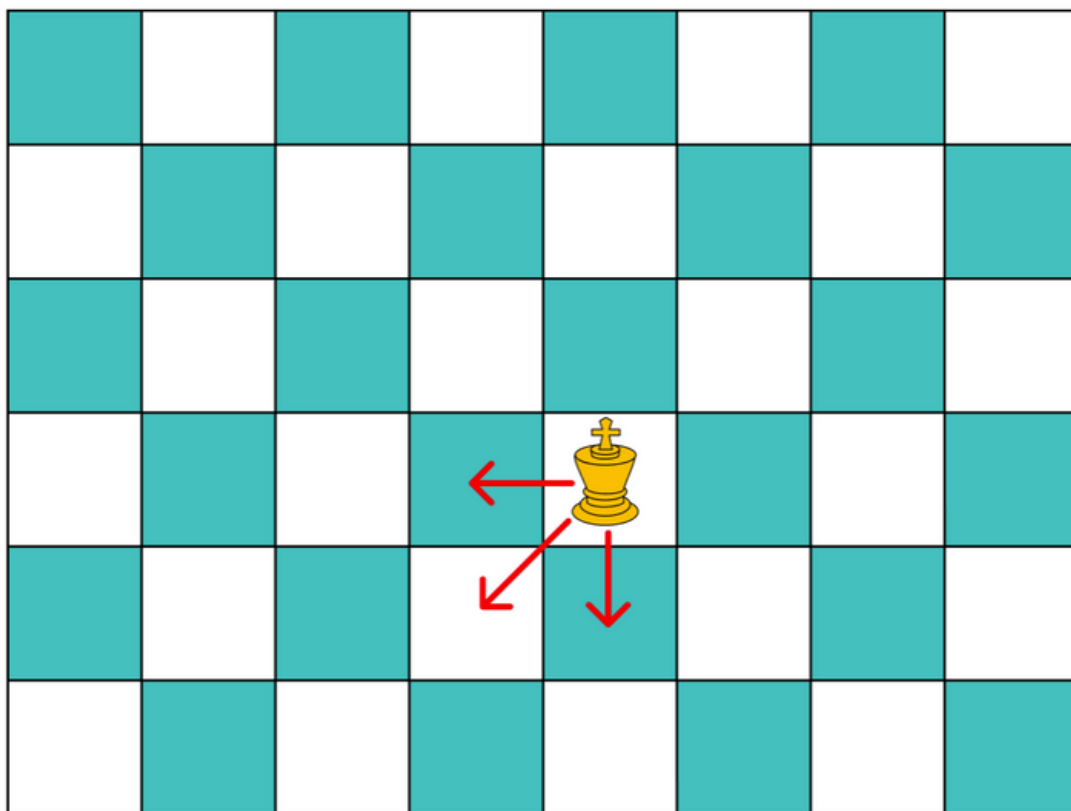
Vector had the moon with him, but he forgot that the effect of shrink ray lasts only an hour. Very soon, the moon begins to grow back to its original size and it starts rolling around inside Vector's ship, hurting him and freeing the girls. The girls see Gru outside the ship, and Gru tells them to jump over to him to save them. Edith and Agnes successfully make it, but before Margo can jump, Vector grabs her. The moon rolls again inside the ship, and knocks Vector over, freeing Margo from Vector.

2. [INTEGER TYPE ANSWER]

Vector and Gru both try to reach Margo who's standing at the a1 position of 128 x 128 floor (board size not decided yet 128x128 is fine for now) But only one person can move on that floor hence they kept a middleman.

Vector and Gru get alternate turns to give directions to the middleman. He(middleman) can take only one step in one chance that too only in 3 directions either South, West or South-West.

If both of them give optimal directions, Gru would start first and is allowed to start from anywhere on the floor (except the winning square obviously). How many squares are winning squares for Gru?



Vector had the moon with him, but he forgot that the effect of shrink ray lasts only an hour. Very soon, the moon begins to grow back to its original size and it starts rolling around inside Vector's ship, hurting him and freeing the girls. The girls see Gru outside the ship, and Gru tells them to jump over to him to save them. Edith and Agnes successfully make it, but before Margo can jump, Vector grabs her.

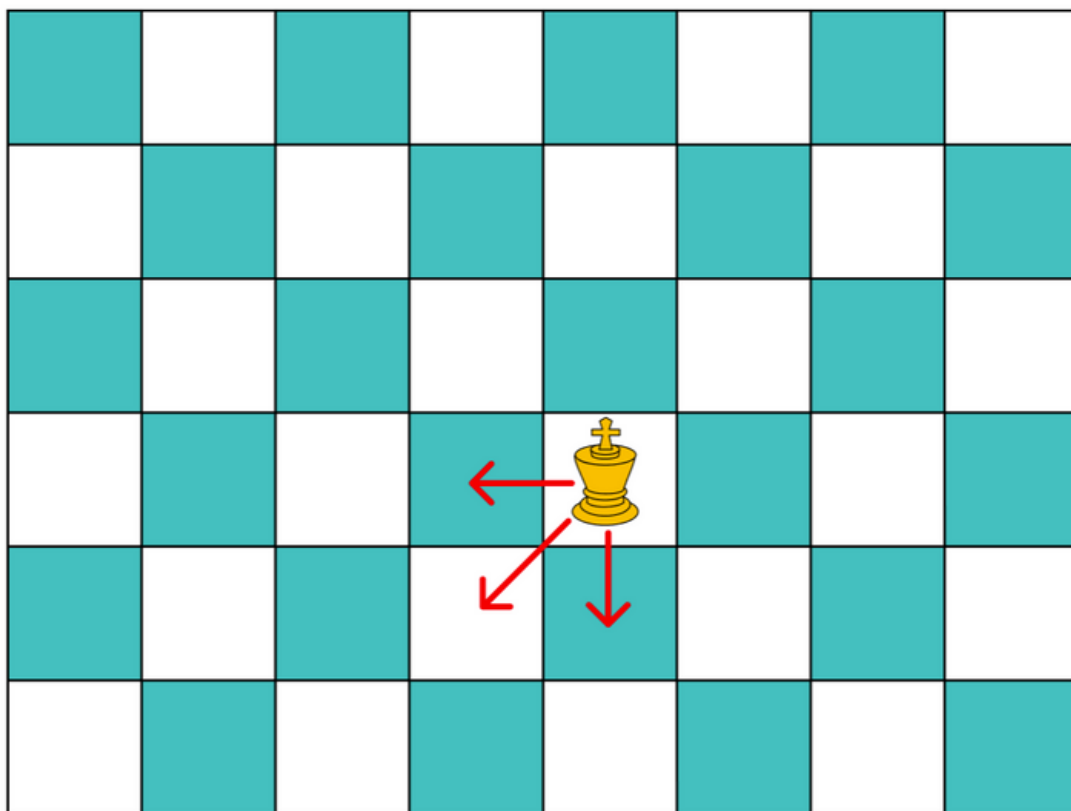
The moon rolls again inside the ship, and knocks Vector over, freeing Margo from Vector.

2. [INTEGER TYPE ANSWER]

Vector and Gru both try to reach Margo who's standing at the a1 position of 128 x 128 floor (board size not decided yet 128x128 is fine for now) But only one person can move on that floor hence they kept a middleman.

Vector and Gru get alternate turns to give directions to the middleman. He(middleman) can take only one step in one chance that too only in 3 directions either South, West or South-West.

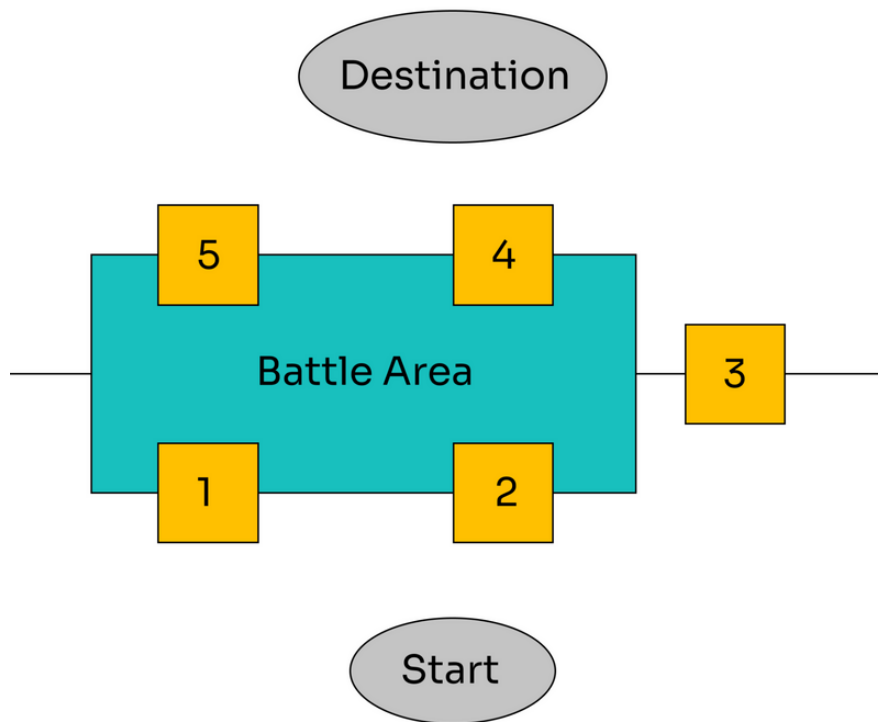
If both of them give optimal directions, Gru would start first and is allowed to start from anywhere on the floor (except the winning square obviously). How many squares are winning squares for Gru?



3. [INTEGER TYPE ANSWER]

Meanwhile, Margo goes from the place she's standing(starting position) towards Gru(finish point). There are a few rooms each one she needs to cross once and only once, how many paths are possible for Margo to reach Gru?

Eg: *Start-1-4-5-2-3-Destination* is one of the paths.



4. [INTEGER TYPE ANSWER]

Gru and Vector are two of the strongest villains; currently, both of them have equal strength so it's no use fighting. They visit President, Bank of Evil Mr. Perkins together, who has created 12 potions that increase strength but there's a catch. If Gru chooses a potion with **even strength** the value of the potion gets added to his strength but if he chooses a potion with an **odd strength**, his strength remains unchanged.

Similarly, if Vector chooses a potion with **odd strength**, the value of the potion gets added to his strength but if he chooses a potion with **even strength**, his strength remains unchanged. Both Gru and Vector alternate to choose a potion with Gru going first(since he has pulled the biggest heist of the century by stealing the moon).

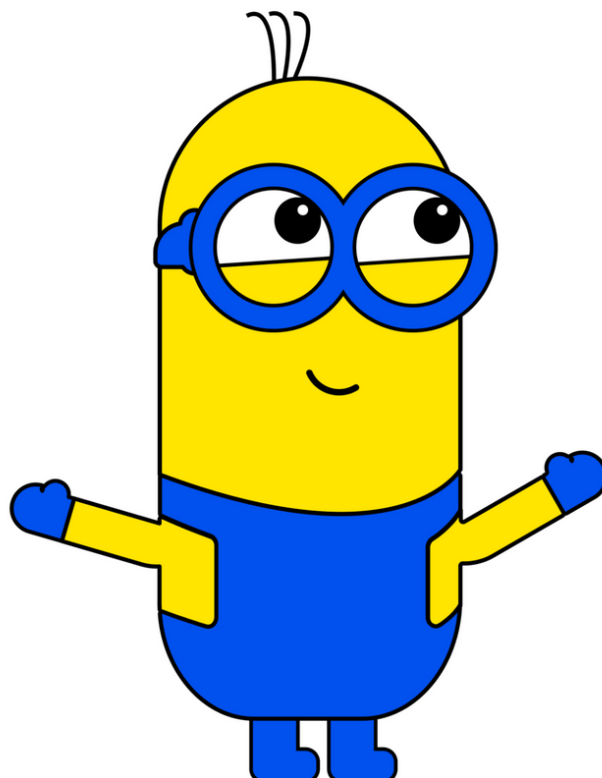
Determine who would be more powerful in the end if both choose their potions optimally.



The strengths of the 12 potions are : 45, 50, 71, 85, 13, 62, 82, 33, 46, 50, 25, 39

- a) Vector
- b) Gru
- c) Both would end up with the same power
- d) It can't be decided.

After pulling out the biggest heist in the history of the world, Gru and the girls settle down to live a happy life as a family. Congratulations to you for helping GRU, minions, and the girls. Throughout this journey, you've completed this journey with the minions; you deserve a hug from us...



Before saying Goodbye, the minions have a small surprise for you i.e "Please insert a very cool name here".

(You can message us this name on our Facebook/Instagram page, hope the page admin is ready for this flood and fury of texts☺)

Solve all these questions correctly and get a bonus of 8 marks.

SECTION 6:

Marking Scheme: All or nothing

Maximum marks: 8

Minimum marks: 0

Vector breaks his promise and keeps the girls and the moon even after GRU solves the puzzle. This enrages Gru, who storms Vector's ship which is conical in shape.

1. [SINGLE CORRECT ANSWER]

Fifteen plus fifteen is equal to x and sixteen added to sixteen is equal to x too. What is the value of x?

- a) THIRTY
- b) TWENTY
- c) TWENTY-FIVE
- d) TWO

2. [INTEGER TYPE ANSWER]

Stuart asks Bob that If you divide 30 by half and add ten, what do you get? And Bob quickly answers it. What was that Quick answer?

3. [SINGLE CORRECT ANSWER]

Which answer in this list is the correct answer?

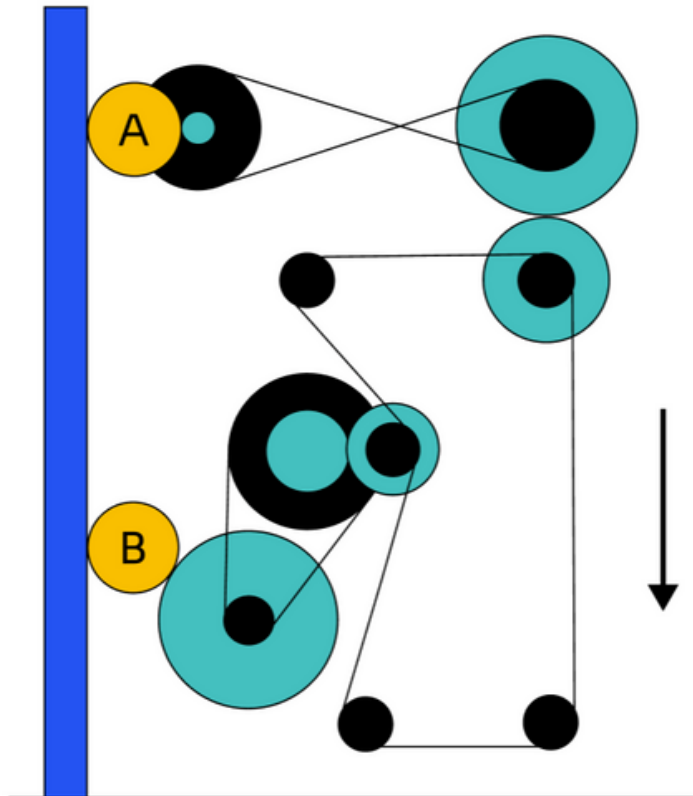
- a) All of the above
- b) None of the below
- c) All of the above
- d) One of the above
- e) None of the above
- f) None of the above

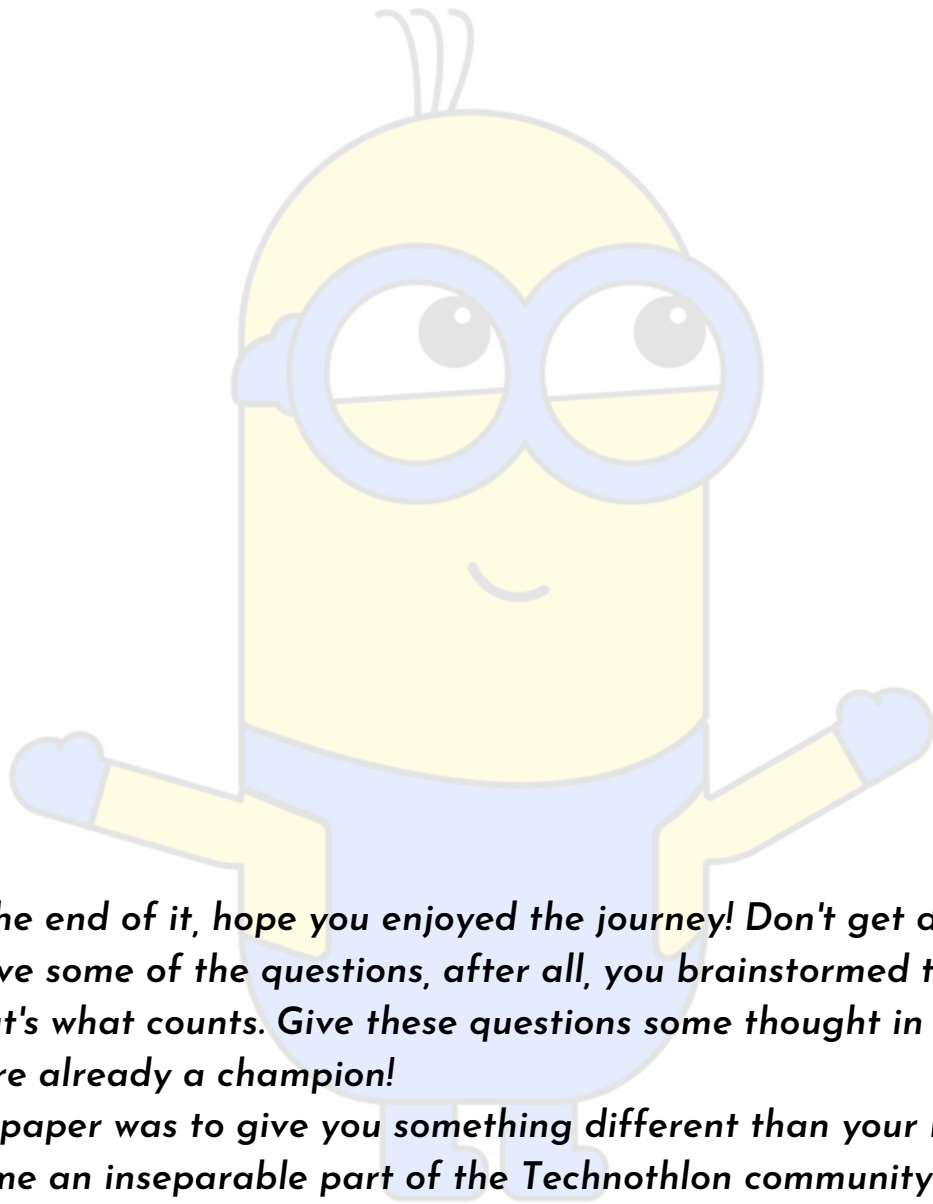
4. [MULTIPLE CORRECT ANSWER]

Bob makes a unique wall-climbing machine as shown in the figure, Stuart said to Bob that “You are a screw head and this design will not work. ” but Bob was adamant that it will. What do you think about it?

1.e. Would the machine work? And if yes in which direction would wheel A and wheel B rotate.

- a) The machine won't work
- b) A will rotate in a clockwise direction.
- c) B will rotate in an anti-clockwise direction.
- d) A will rotate in an anti-clockwise direction
- e) B will rotate in a clockwise direction.





Well that was the end of it, hope you enjoyed the journey! Don't get disheartened if you couldn't solve some of the questions, after all, you brainstormed through these puzzles and that's what counts. Give these questions some thought in your leisure time too. You are already a champion!

The aim of this paper was to give you something different than your routine "exam". You have become an inseparable part of the Technothlon community. Hope to see you again next year!

Feedback is always appreciated.

TEAM TECHNOHLON 2021